

Unified Military Analytics and Comparison Dashboard-DV

Report for Milestone 2

**INTERNSHIP OF
DATA VISUALIZATION
IN
INFOSYS SPRINGBOARD**

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ABSTRACT

This project aims to develop a fully interactive dashboard suite for analyzing global military power in 2025. Using Python, military metrics will be scraped from GlobalFirepower.com for 140+ countries via a pre-defined list (links_for_military_data.txt).

The project leverages Python-based data pipelines to scrape, clean, and process open-source defense data from GlobalFirepower.com. Unlike traditional single-platform dashboards, GDM emphasizes cross-platform flexibility — allowing deployment in tools like Tableau, Power BI, or Python-based web apps built using Streamlit or Dash.

Dashboards planned:

- Quick Stats – A global overview of rankings, trends, and highlights.
- Nation Overview – Detailed analysis of individual country capabilities.
- Compare Powers – Side-by-side military and economic comparison.
- Coalition Builder – Interactive simulation of alliance strength and combined assets.

The final deliverable is a comprehensive analytics suite that enables users to compare nations, evaluate coalitions, and analyze KPIs like Power Index Rank Gap, Assets per Capita, and Defense Budget-to-GDP Ratio through intuitive and dynamic visualizations.

Milestone 2: KPI Engineering and Dashboard Prep

Module 3: KPI Feature Engineering

3.1 Methodologies:

3.1.1 Development Environment:

Google Collab: cloud based platform that can be used for python development, without need of any local installation.

3.1.2 Data Source:

military_cleaned.csv

3.1.3 Tools and libraries:

The tools and libraries used for completing this milestone are:

- i. **Requests:** used to send the Http requests and also to fetch the content from the web page.
- ii. **BeautifulSoup (bs4):** used for parsing and extracting the HTML elements
- iii. **pandas:** used for structuring, storing the scraped data, Data cleaning, aggregation, and KPI calculations.
- iv. **NumPy:** Numerical and ratio-based computations

3.2 Implementation steps:

i. Load Required Libraries

- Import pandas and numpy for data processing.
- Import requests and BeautifulSoup for external metadata.

ii. Load and Prepare Dataset

- Load military_cleaned.csv dataset using Pandas.

- Standardize column names into lowercase.

iii. Calculate the KPI's:

1. Power Index Rank Gap:

- This metric will help to show how far a nation lags or leads compared to top military powers. So for this first calculate ranks of each country with the help of power index, were the one with lower power index has stronger military and also rank according to total population.
- Formula: $\text{power_index_rank_gap} = \text{population_rank} - \text{power_rank}$

2. Assets per Capita:

- This will help to measure the total assets available relative to the population size.
- Formula: $\text{assets_per_capita} = (\text{total_military_aircraft} + \text{tanks} + \text{total_naval_fleet} + \text{total_military_helicopters}) / \text{total_population}$

3. budget_to_gdp_ratio:

- This shows how much of a country's economy (GDP) is spent on its military budget.
- Formula: $\text{budget_to_gdp_ratio} = \text{defense_budget_usd}] / \text{df[\"purchasing_power_parity_usd\"}$

4. Budget per soldier:

- Indicates how much financial support each service member gets.
- Formula: $\text{budget_per_soldier} = \text{defense_budget_usd} / \text{total_military_manpower}$

5. air_power_density:

- Shows concentration of aircraft relative to population.
- Formula: $\text{air_power_density} = \text{total_military_aircraft} / \text{total_population}$.

6. military_participation_ratio:

- This shows Reveals proportion of citizens engaged in active military service.
- Formula: $\text{military_participation_ratio} = \text{total_military_manpower} / \text{total_population}$

iii. Handle Infinite & Null Values

- Replace infinite values with NaN.
- Ensure clean numeric outputs for dashboards.

iv. Alliance (NATO) Feature Engineering

- Scrape NATO member countries from website “<https://www.nato.int/en/about-us/organization/nato-member-countries>”, and store in nato_countries.
- Create binary flag is_nato (1 = NATO, 0 = Non-NATO).

v. Region & Continent Mapping

- Scrape country–region–continent mapping from "<https://statisticstimes.com/geography/countries-by-continents.php>"
- Map region and continent to each country.
- Then Manually fix unmatched country names.

viii. Export Final Dataset

- Save engineered KPIs and metadata to final CSV.
- Save it as military_final.csv

3.3 Outcomes:

AutoSaveOff

military_final

Search

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelp

Cut

Paste

Copy

Format Painter

Clipboard

Calibri

11

A⁺

A⁻

B

I

U

Font

Alignment

Wrap Text

General

Number

Conditional Formatting

Styles

Cells

Σ AutoSum

Fill

Clear

Editing

BE

BF

BG

BH

BI

BJ

BK

BL

BM

BN

BO

power_rank

population_rank

power_index_rank

assets_per_capita

budget_to_gdp_ratio

budget_per_soldier

air_power_density

military_participation_ratio

is_nato

region

continent

1

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6

7

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9

10

11

12

13

14

15

16

17

18

19

20

21

22

7.01E-05

8.60E-05

8.32E-06

5.41E-06

9.34E-05

1.80E-05

2.59E-05

2.21E-05

4.77E-05

2.49E-05

4.83E-06

1.79E-05

4.74E-06

1.49E-05

0.000225469

2.83E-05

2.29E-05

1.86E-05

4.68E-05

4.66E-05

3.54E-05

0.036234818

0.021649485

0.008557692

0.005725191

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0.01932447

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0.015986395

0.009964516

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0.002710997

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0.010729167

0.010445089

0.035253165

0.003078272

0.095892857

0.030061728

5948.270648

1826.028815

349.4200176

113.2433921

1777.972344

2270.495233

1828.164503

1075.941946

1093.402641

1125.966079

233.0764425

70.40414634

76.83074176

1292.163756

7723.280677

312.1421162

1075.773648

4954.277004

128.9042566

2952.572287

2565.089542

3.81E-05

3.05E-05

2.34E-06

1.58E-06

3.06E-05

9.22E-06

1.43E-05

1.17E-05

1.29E-05

1.20E-05

2.33E-06

5.54E-06

1.63E-06

6.94E-06

6.50E-05

6.23E-06

9.75E-06

1.22E-05

9.82E-06

9.09E-06

1.24E-05

0.440000001

0.490000001

0.54

0.47

0.500000001

0.459999996

0.439999999

0.429999997

0.510999996

0.450000001

0.509999999

0.430000002

0.490000001

0.46

0.419999985

0.560000003

0.459999996

0.419999994

0.410000003

0.509999993

0.490000003

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1 Northern America

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0 Eastern Asia

0 Western Asia

1 Southern Europe

0 South America

0 Southern Asia

0 South-eastern Asia

1 Western Europe

0 Western Asia

0 Southern Asia

1 Southern Europe

0 Australia and New Zealand

0 Northern Africa

0 Eastern Europe

1 Eastern Europe

North America

Europe

Asia

Asia

Europe

Europe

Europe

Asia

Asia

Europe

South America

Asia

Asia

Europe

Europe

Asia

Europe

Oceania

Africa

Europe

Europe

Figure 1: snapshot of military_final.csv

Module 4: Dashboard Planning and Prototyping

4.1 Tools Used:

- Figma – for designing low-fidelity wireframes and storyboard layouts
- Microsoft Power BI – for building the interactive operational dashboard using sample data

4.2 Dataset:

- military_final.csv

4.3 Implementation:

4.3.1 Figma wireframe for story board

Draft layout sketches or wireframes for:

- o Quick Stats
- o Nation Overview
- o Compare Powers
- o Coalition Builder
- In the figma, create the wireframe for the 4 dashboards.
- The navigation between the dashboard will happen with the help of the buttons.
- So sketch the buttons Quick Stats, Nation Overview, Compare Powers and Coalition Builder.

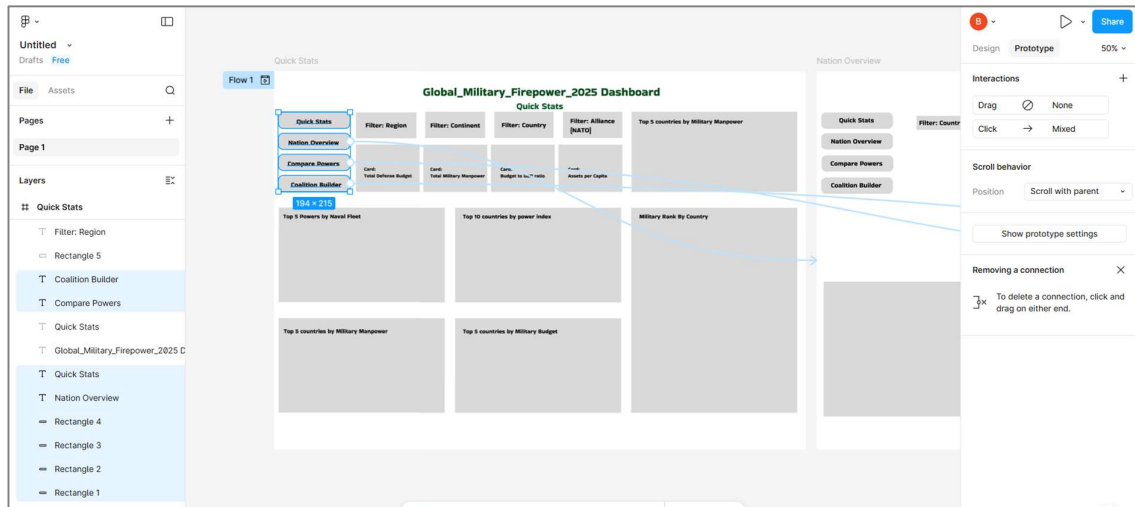


Figure 2. Wireframe for the dashboard.

- This image shows the flow of the dashboard from one to another when the button is clicked.
- Filters:

Set filters on the basis of Region, Continent, Country and Alliance (NATO).

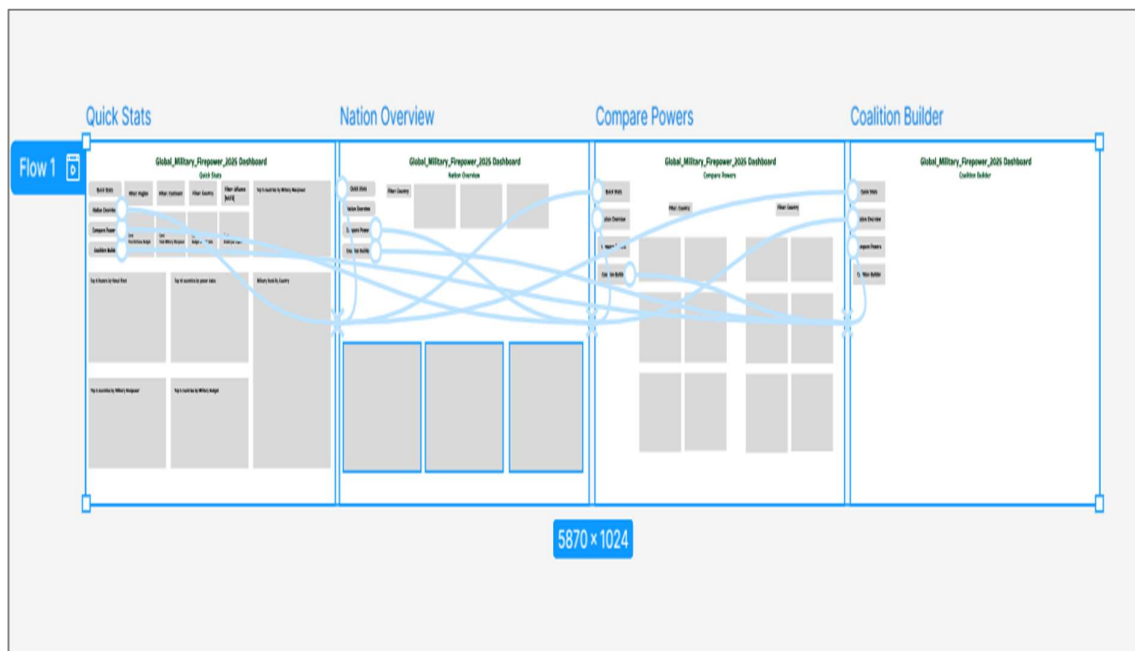


Figure 3. Overall flow of the dashboard planned using wireframe.

4.3.2 Power Bi

- Create global_power_2025.pbix
- First import the military_final.csv.
- Create 4 blank dashboards:
- Quick stats, Nation Overview, Compare Powers Coalition Builder.
- Insert Page Navigator Buttons in all 4 dashboards. We can alt+click to test if it navigates correctly.
- Next, sample data is plotted in the first dashboard Quick Stats, to check if it working properly.
- In quick stats, using slicer, filters are added. Filters: Region, Continent, Country and Alliance.
- For showing important KPI's cards are used. In cards, Total Defense Budget (USD) , Total Military Manpower, Budget to GDP Ratio and Assets per Capita is visualized
- For quick stats:
 - Top 5 countries by military power is visualized using Pie chart using columns country and total_manpower and applying filter Top N.
 - Top 5 Powers by Naval fleet is visualized using Donut Chart, using columns country and total_naval_fleet, and applying filter Top N.
 - Top 10 Countries by Power Index is visualized using Funnel where lower values is stronger country, using columns country and pwr_index, and in filters Top 10, but from bottom.

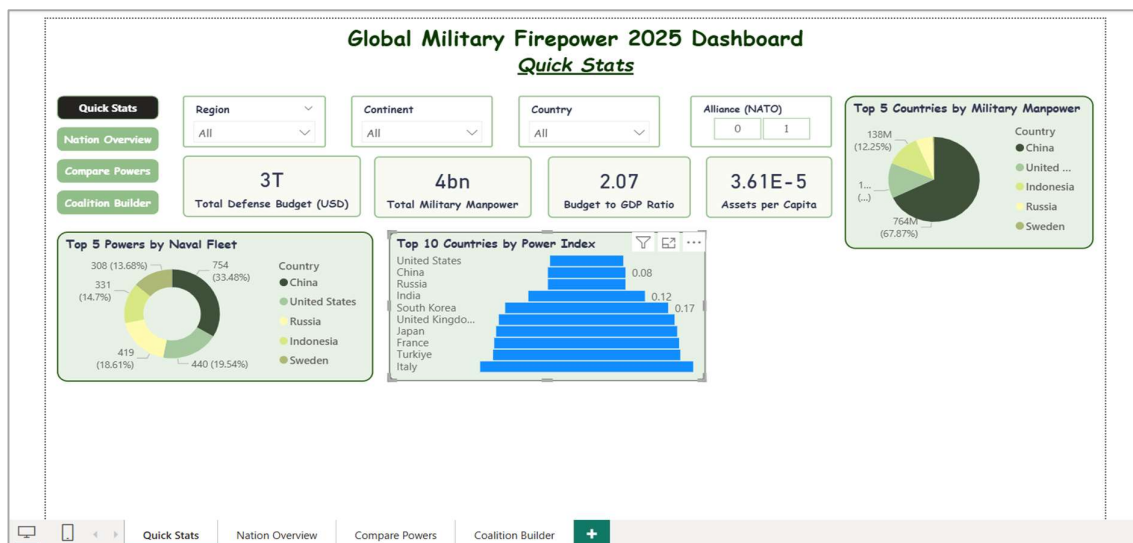


Figure 4. Quick Stats Dashboard (Operational)

CONCLUSION

This concludes the report for milestone 2 including the tools used, planning done for dashboards, outcomes of the KPI engineering and Dashboard Planning and Prototyping, completing the second milestone of the data visualization internship of Infosys Springboard.

Future works include full dashboard development.